

Explain why a computer system requires an Operating System (OS)

9618_s25_qp_11_sc_6.b

QUESTION

(b) Both computers have an Operating System (OS).

Describe the purpose of an OS in a computer.

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..... [5]

MARK SCHEME

6(b)	1 mark each to max 5 e.g. • To provide a user interface • so that the user is able to communicate with the hardware • To manage memory • so that data can be stored and accessed • and multitasking is possible • To manage files • allowing the user to create, edit, update and delete files and folders • To manage inputs and outputs from hardware/peripherals • To handle processes • to make sure each process has fair access	5
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QUESTION

8 A computer has an Operating System (OS).

(a) State **one** purpose of the Operating System.

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 [1]

MARK SCHEME

8(a)	1 mark for each bullet point (max 1) • To hide the complexities of the hardware from the user • To provide a platform for software to run • To provide a user interface	1
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Explain the key management tasks carried out by the Operating System

9618_w25_qp_13_sc_8.d

QUESTION

(d) Memory management is one key management task performed by an OS.

Give **three** other key management tasks that are performed by an OS.

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2

3 [3]

MARK SCHEME

8(d)	1 mark for each bullet point, max 3 marks • File management • Security management • Hardware / peripheral management // input / output management • Process management	3
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9618_w25_qp_11_sc_5.a.i

QUESTION

5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(a) (i) State **two** tasks that are performed by hardware management.

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..... [2]

MARK SCHEME

5(a)(i)	1 mark per bullet point, max 2 marks • Installs <u>driver</u> software for devices connected to external ports • Manages communication between devices • Manages <u>hardware</u> interrupts	2
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9618_w25_qp_11_sc_5.a.ii

QUESTION

(ii) State **two** tasks that are performed by security management.

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[2]

MARK SCHEME

5(a)(ii)	1 mark per bullet point, max 2 marks • Prevents unauthorised access • ... by providing authentication // by validating users and processes • Implements access rights and permissions • Makes provision for recovery of lost data • Carries out Operating System security updates as available • Carries out auditing and keeps logs of activity	2
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[5]

6(b)	1 mark each to max 5 e.g. • To provide a user interface • so that the user is able to communicate with the hardware • To manage memory • so that data can be stored and accessed • and multitasking is possible • To manage files • allowing the user to create, edit, update and delete files and folders • To manage inputs and outputs from hardware/peripherals • To handle processes • to make sure each process has fair access	5
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QUESTION

- (c) The computer has an Operating System (OS). One of the key management tasks of the OS is process management.

Describe the process management tasks performed by an OS.

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MARK SCHEME

3(c)	1 mark for each bullet point (max 4) e.g. • It manages the scheduling of processes // Decides which process is to be run next • Allows multi-tasking/multi-processing • Ensures fair access • Handles interrupts • Manages / allocates which resources the processes require • Facilitates the sharing and exchange of data between processes • Prevents interference between processes // conflict resolution	4
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QUESTION

- 4** A student uses a laptop to write a program that is saved as a text file.

- (a) The laptop has utility software and an Operating System (OS).

- (i) Describe the file management tasks carried out by an OS.

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..... [2]

MARK SCHEME

4(a)(i)	1 mark for each bullet point (max 2) e.g. • Secondary storage space is divided into file allocation units • Space on secondary storage is allocated to particular files • Maintains/creates directory structures • Specifies the logical method of file storage e.g. FAT or NTFS • Provides file naming conventions • Controls user access // implements access rights //implements password protection // Makes file sharing possible • Controls access to the file by other software • Specifies tasks that can be performed on a file e.g. open, close, delete, copy, create, move, etc • Allows searching for a file	2
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9618_s24_qp_12_sc_6

QUESTION

6 A computer has an Operating System (OS).

Memory management and process management are two OS tasks.

Explain how memory management **and** process management support multi-tasking.

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MARK SCHEME

6	1 mark each to max 4 Max 3 marks for each management task: <i>Memory management: Max 3 marks</i> • Stores data from all currently running programs concurrently in RAM • Stops the data from overwriting each other in RAM/primary storage • Decides which processes should be in main memory • Makes efficient use of memory <i>Process management: Max 3 marks</i> • Allows one process to be paused whilst another process can be actioned • Decides which process is to be run next • Switches between processes to allow them to share the use of the processor • Identification/description of scheduling	4
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9618_s23_qp_12_sc_5.d.i

QUESTION

(d) The laptop has systems software.

(i) Describe how the Operating System (OS) manages processes in the computer.

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..... [5]

MARK SCHEME

5(d)(i)	1 mark each to max 5 • Manages the scheduling of processes // decides which order to run processes • Manages which resources the processes require • ... such as allocating memory • Enables processes to share data • Prevents interference between processes // resolution of conflicts • Handles the process queue • It allows multi-tasking / multi-processing • ... by ensuring fair access, handling priorities and handling interrupts	5
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9618_s23_qp_11_sc_3.a

QUESTION

3 A computer has an Operating System (OS).

(a) Describe how the Operating System manages the peripheral hardware devices of the computer.

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MARK SCHEME

3(a)	1 mark each to max 4 Examples: • Installs device drivers • ... to allow communication between peripherals and computer • Sends data and receives data to and from peripherals • ... such as to an output device and from an input device/by example • Handles buffers for transfer of data • ... to ensure smooth transfer between devices that transmit and receive at different speeds • Manages interrupts / signals from the device	4
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QUESTION

(b) Hardware management is one key management task carried out by the Operating System.

Identify **two other** key management tasks carried out by the Operating System.

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2 [2]

MARK SCHEME

3(b)	1 mark each to max 2 • Memory management • File management • Security management • Process management • Error checking and recovery	2
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QUESTION

3 Draw **one** line from each Operating System (OS) management task to its most appropriate description.

OS Management task

Description

hardware management

dynamically allocates memory to processes

security management

marks unallocated file storage for availability

memory management

installs programs for devices connected to external ports

process management

validates user and process authenticity

allows processes to transfer data to and from each other

[4]

MARK SCHEME

3	1 mark for each correct line: <table><thead><tr><th>OS Management task</th><th>Description</th></tr></thead><tbody><tr><td>hardware management</td><td>dynamically allocates memory to processes</td></tr><tr><td>security management</td><td>marks unallocated file storage for availability</td></tr><tr><td>memory management</td><td>installs programs for devices connected to external ports</td></tr><tr><td>process management</td><td>validates user and process authenticity</td></tr><tr><td></td><td>allows processes to transfer data to and from each other</td></tr></tbody></table>	OS Management task	Description	hardware management	dynamically allocates memory to processes	security management	marks unallocated file storage for availability	memory management	installs programs for devices connected to external ports	process management	validates user and process authenticity		allows processes to transfer data to and from each other	4
OS Management task	Description													
hardware management	dynamically allocates memory to processes													
security management	marks unallocated file storage for availability													
memory management	installs programs for devices connected to external ports													
process management	validates user and process authenticity													
	allows processes to transfer data to and from each other													

9618_w22_qp_11_sc_7.b

QUESTION

(b) Memory management is one of the tasks performed by an Operating System (OS).

Describe the ways in which memory management organises and allocates Random Access Memory (RAM).

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..... [2]

MARK SCHEME

7(b)	1 mark for each bullet point (max 2): • RAM is assigned into blocks • dynamic allocation of RAM to programs / processes • reclaims unused blocks of RAM • prevents two programs / processes occupying the same area of RAM at the same time • moves data from secondary storage when needed // manages paging, segmentation and virtual memory	2
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9618_w21_qp_12_sc_7.b

QUESTION

(b) Describe the file management tasks that an Operating System performs.

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..... [4]

MARK SCHEME

7(b)	1 mark for each bullet point to max 4 • Storage space is divided into file allocation units • Space is allocated to particular files • Maintains / creates directory structures • Specifies the logical method of file storage (e.g. FAT or NTFS) • Provides file naming conventions • Controls access // implements access rights // implements password protection // Makes file sharing possible • Specifies tasks that can be performed on a file (e.g. open, close, delete, copy, create, move etc.)	4
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QUESTION

(b) Identify **four** key management tasks that the Operating System will perform.

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- 2
- 3
- 4

[4]

MARK SCHEME

2(b)	1 mark per bullet point to max 4 • memory management • file management • security management • hardware / device / peripheral / resources management • input/output management • process management • error checking and recovery • provision of a platform for software • provision of a user interface	4
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Show understanding of the need for typical utility software provided with an Operating System

9618_w25_qp_12_sc_8.b.ii

QUESTION

(ii) The image files that are displayed on the touchscreen have been corrupted.

Identify **two** examples of utility software that could be used to recover the images.

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2 [2]

MARK SCHEME

8(b)(ii)	1 mark per bullet point, max 2 marks • Disk repair / disk contents analysis • Back-up software	2
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9618_w24_qp_11_sc_4.a.ii

QUESTION

(ii) Explain the need for back-up software.

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..... [2]

MARK SCHEME

4(a)(ii)	1 mark for each bullet point (max 2) • To allow data to be retrieved / restored when lost • To automatically make a duplicate copy of data... • ...so the user does not have to remember to backup data • To make regular duplicate copies of data	2
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QUESTION

- (ii) The program files are stored on a new hard disk after they have been downloaded.

Describe the reasons why a hard disk formatter is needed for the new hard disk.

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MARK SCHEME

5(b)(ii)	1 mark for each bullet point (max 3). • Disk needs to be prepared for initial use • Disk needs to be checked for errors • A new file system needs to be generated on the disk • The file allocation table needs to be set up	3
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QUESTION

- (b) The Operating System has utility software.

The purpose of some utility software is to improve security.

Identify **one** example of utility software that is **not** intended to improve security.

Explain why this software is needed.

Utility software

Explanation

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..... [3]

MARK SCHEME

8(b)	1 mark for the name of the utility software 2 marks for the explanation • Defragmentation software • ... because over time saving and deleting of small files fragments the disk • ... the software makes (individual) files contiguous • ... so access time to the files is improved • ... because head movement is reduced • Disk contents analysis/disk repair software • ... to identify and mark bad sectors • ... to restore corrupted files • ... to recover lost data (due to hardware failure) • File compression • ... to reduce the size of files • ... which saves storage and memory space // by example • ... and reduces transmission time // by example • Disk formatter • ... to prepare a disk for use // set up the file system • ... to partition the disc • ... to delete all the data from the disc	3
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9618_s23_qp_12_sc_5.d.ii

QUESTION

(ii) Describe the purpose of utility software in a computer.

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MARK SCHEME

5(d)(ii)	1 mark each to max 2 • To help users to set-up / configure / analyse / optimise / maintain the computer ... • ... by for example, making memory allocation more efficient • ... by for example, checking the system for faults	2
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QUESTION

(c) The Operating System has utility software including defragmentation software.

Explain how defragmentation can improve the performance of the computer.

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MARK SCHEME

3(c)	1 mark each to max 3 • Rearranges blocks of individual files (on the HDD) so they are contiguous // moves the free space together • Accessing each file is faster • ...because there is no need to search for the next fragment / block of the file • ...so less head movement is needed	3
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QUESTION

- 1 (a) Draw **one** line from each utility software to its most appropriate purpose.

Utility software

virus checker

disk formatter

backup

disk repair

defragmentation

Purpose

to reorganise files so
they are contiguousto scan for malicious
program code

to decrease the file size

to initialise a disk

to create copies of files
in case the original is lostto check for and fix
inconsistencies on a disk

[5]

MARK SCHEME

1(a)	1 mark for each correct line. <table><tr><th>Utility software</th><th>Purpose</th></tr><tr><td>virus checker</td><td>to reorganise files so they are contiguous</td></tr><tr><td>disk formatter</td><td>to scan for malicious program code</td></tr><tr><td>backup</td><td>to decrease the file size</td></tr><tr><td>disk repair</td><td>to initialise a disk</td></tr><tr><td>defragmentation</td><td>to create copies of files in case the original is lost</td></tr><tr><td></td><td>to check for and fix inconsistencies on a disk</td></tr></table>	Utility software	Purpose	virus checker	to reorganise files so they are contiguous	disk formatter	to scan for malicious program code	backup	to decrease the file size	disk repair	to initialise a disk	defragmentation	to create copies of files in case the original is lost		to check for and fix inconsistencies on a disk	5
Utility software	Purpose															
virus checker	to reorganise files so they are contiguous															
disk formatter	to scan for malicious program code															
backup	to decrease the file size															
disk repair	to initialise a disk															
defragmentation	to create copies of files in case the original is lost															
	to check for and fix inconsistencies on a disk															

9618_w21_qp_12_sc_7.c

QUESTION

- (c) Identify **two** utility programs that can be used to improve the performance of a computer **and** state how they improve the performance.

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[4]

MARK SCHEME

7(c)	1 mark for identifying program 1 mark for description, max 2 per program e.g. • Defragmentation • Less time is taken to access files because each one is contiguous so there is less head movement • Virus checker • makes more RAM available for programs to run • ... because it removes software that might be taking up memory / replicating • Disk repair / Disk contents analysis • preventing bad sectors being used because it identifies / marks them • reduces access times by optimising storage • Disk/system clean up • releases storage by removing unwanted / temporary files	4
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QUESTION

2 Bingwen's computer comes with an Operating System and utility software.

(a) Draw **one** line from each utility software to its correct description.

Utility software

Description

Disk formatter

Scans software for errors and repairs the problems

Defragmentation

Moves parts of files so that each file is contiguous in memory

Back-up

Creates a copy of data that is no longer required

Disk repair

Sets up a disk so it is ready to store files

Scans for errors in a disk and corrects them

Creates a copy of data in case the original is lost

[4]

MARK SCHEME

2(a)

1 mark for each correct line

4

Utility software

Description

Disk formatter

Scans software for errors and repairs the problems

Defragmentation

Moves parts of files so that each file is contiguous in memory

Back-up

Creates a copy of data that is no longer required

Disk repair

Sets up a disk so it is ready to store files

Scans for errors in a disk and corrects them

Creates a copy of data in case the original is lost

Show understanding of program libraries

9618_w25_qp_13_sc_8.b

QUESTION

(b) Explain how a programmer benefits from using program libraries during program development.

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..... [3]

MARK SCHEME

8(b)	1 mark per bullet point, max 3 marks	3
	• Programming time is saved as code does not have to be written from scratch • Testing time is saved as code is already tested / documented • A library routine is more likely to work, as code is already tested • Library routines automatically update if they are changed / improved • The programmer can use library routines to perform complex functions / procedures that they may not be able to write themselves	

9618_w25_qp_11_sc_5.b.i

QUESTION

(b) (i) Describe **one** benefit of using library routines when developing software.

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..... [2]

MARK SCHEME

5(b)(i)	One mark for benefit, one mark for appropriate matching expansion e.g. • Less code needs to be developed as code is pre-written • ... thereby reducing the time to write a program • The library routine will be pre-tested and documented • ... reducing the time taken to test a program // giving confidence that the routine works as it should • The code for the library routine can be written in a different programming language • ... which allows the use of special features of that language • The user's source code is more readable • ... as the library routine is simply called in the source code rather than being written out in full • The library could contain the code for complex routines • ... thus allowing the programmer to include modules they might not be able to code	2
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9618_w25_qp_11_sc_5.b.ii

QUESTION

(ii) Describe **one** drawback of using library routines when developing software.

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..... [2]

MARK SCHEME

5(b)(ii)	1 mark for drawback, 1 mark for appropriate matching expansion e.g. • Compatibility issues • ... the library routine may not work with current program • Not guaranteed to be thoroughly tested • ... there could be unexpected problems (bugs/virus) • The library routine may not match needs exactly // The library routine may give unexpected results • ... so it will need to be edited and tested • If the library routine is later edited or changed (by someone else) • ... there could be unexpected errors or results	2
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9618_w24_qp_12_sc_4.b

QUESTION

(b) The programmers created a new program library whilst developing the program.

Describe the benefits to the programmers of creating a program library.

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..... [3]

MARK SCHEME

4(b)	1 mark for each bullet point (max 3) e.g. • Subroutines can be shared / reused • ... between team members who are working independently • ... without having to rewrite/re-test them which saves the programmers' time • A program library provides continuity between programs/programmers • Individual programmers can contribute their specialisms to the library // Individual programmers can use the specialisms of others	3
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QUESTION

(c) The student used a program library when writing their program.

Explain the benefits to the student of using library files when writing a program.

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..... [3]

MARK SCHEME

4(c)	1 mark for each bullet point (max 3) e.g. • It saves development time for the student • ... by using pre-written code • There is less testing to do // the routines are more likely to work correctly • ... because they have already been tested and are error-free • It results in a more robust program • Subroutines can be used that they do not have the skill to write • ... which enables a more complex program to be written	3
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QUESTION

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(e) Program libraries were used when writing the robot's software.

(i) State what is meant by a **program library**.

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MARK SCHEME

7(e)(i)	1 mark for: Pre-written code/functions/routines that can be imported/called in another program	1
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QUESTION

- (ii) Some program libraries include Dynamic Link Library (DLL) files.

Describe the benefits of a programmer using a library with DLL files instead of using a library that does not include DLL files.

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..... [4]

MARK SCHEME

7(e)(ii)	1 mark each to max 4: • Maintenance not needed to be done by the programmer • ... because the DLL is separate from program • The calling program does not need recompilation by the programmer when a DLL file changes • ... because the DLL file can be updated independently of the calling program • ... updates will apply to all programs that use the DLL file	4
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QUESTION

- (c) The programmer uses program libraries when developing the program.

Describe **two** benefits to the programmer of using program libraries.

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2

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[2]

MARK SCHEME

8(c)	1 mark each to max 2: • Saves programming/testing time as code does not have to be written/re-written from scratch // code does not have to be tested • Code is already tested so it is more robust/likely to work • The programmer does not need to maintain the library // library routines are updated automatically • Can perform complex calculations that the programmer may be unable to do • Makes code more easily readable	2
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9618_s23_qp_12_sc_7.a.i

QUESTION

7 A software developer is working in a team writing a program for a client.

(a) The developer is writing a new program library to be used by the other team members.

(i) Define the term **program library**.

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..... [2]

MARK SCHEME

7(a)(i)	1 mark each to max 2 • Set of pre-written / pre-compiled / pre-tested subroutines • ... which can be called in other programs • ... by installing/importing the library	2
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QUESTION

(ii) Explain **two** benefits to the developer of choosing to create a Dynamic Link Library (DLL).

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[4]

MARK SCHEME

7(a)(ii)	1 mark for each bullet point. Mark in pairs; 1 mark for a benefit and 1 mark for an appropriate expansion • (main) memory requirements for program is reduced • ... as dynamic link library is loaded only once / when required • the executable file size of the program using the DLL will be smaller • ... because the executable does not contain (all) the library routines • maintenance not needed to be done by the programmer • ... because the DLL is separate from program • no need to recompile the main program when changes are made to DLL • ... because changes / improvements/ error correction to the DLL file code are done independently of the main program • A single DLL file can be made available to several application programs • ... Saving space in memory / easing the pressure on memory	4
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QUESTION

7 (a) State **two** benefits to a programmer of using Dynamic Link Library (DLL) files.

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- 2
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- [2]

MARK SCHEME

7(a)	1 mark for each benefit (max 2): • (main) memory requirements for program are reduced as dynamic link library is loaded only once / when required • the executable file size is smaller because the executable does not contain all the library routines • maintenance not needed to be done by the programmer because the DLL is separate from program • no need to recompile the main program when changes are made to DLL because changes / improvements/ error correction to the DLL file code are done independently of the main program	2
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QUESTION

7 Jennifer is writing a computer program for her A Level homework.

(a) Jennifer uses a program library to help her write her computer program.

Describe how a program library can be used while writing a computer program.

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- [2]

MARK SCHEME

7(a)	1 mark per bullet point to max 2 • Program libraries store pre-written functions and routines • The program library can be referenced/imported • the functions/routines can be called in her own program	2
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